

Week_7 Task and hands-on_Activity

Github_repo_url =

https://github.com/horrendor/DevOps_intern_Davine/tree/main/weekly_Tasks/week_7/k-8

Screenshot of Running Cluster

```
horrendor@horrendor-Nitro-AN515-55:~$ kubectl get nodes
NAME                                STATUS    ROLES    AGE   VERSION
devops-control-plane               Ready    control-plane   51m   v1.37.0
horrendor@horrendor-Nitro-AN515-55:~$
```

Screenshot of Pods

```
horrendor@horrendor-Nitro-AN515-55:~$ kubectl get pods -o wide
NAME                                READY    STATUS    RESTARTS   AGE   IP            NODE                                NOMINATED NODE   READINESS GATES
webapp-5ffd64b6bb-2n2l6             1/1      Running   0           25m   10.244.0.9    devops-control-plane               <none>            <none>
webapp-5ffd64b6bb-6rdf5             1/1      Running   0           34m   10.244.0.5    devops-control-plane               <none>            <none>
webapp-5ffd64b6bb-bm4ms             1/1      Running   0           34m   10.244.0.6    devops-control-plane               <none>            <none>
webapp-5ffd64b6bb-gw4qx             1/1      Running   0           34m   10.244.0.7    devops-control-plane               <none>            <none>
webapp-5ffd64b6bb-h2wdq             1/1      Running   0           25m   10.244.0.8    devops-control-plane               <none>            <none>
```

Screenshot of Deployment

```
horrendor@horrendor-Nitro-AN515-55:~$ kubectl get deployment
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
webapp    5/5     5            5           39m
horrendor@horrendor-Nitro-AN515-55:~$
```

Screenshot of Service

```
horrondor@horrondor-Nitro-AN515-55:~$ kubectl get services
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)          AGE
webapp-service      NodePort    10.96.125.45  <none>       80:30587/TCP     35m
horrondor@horrondor-Nitro-AN515-55:~$
```

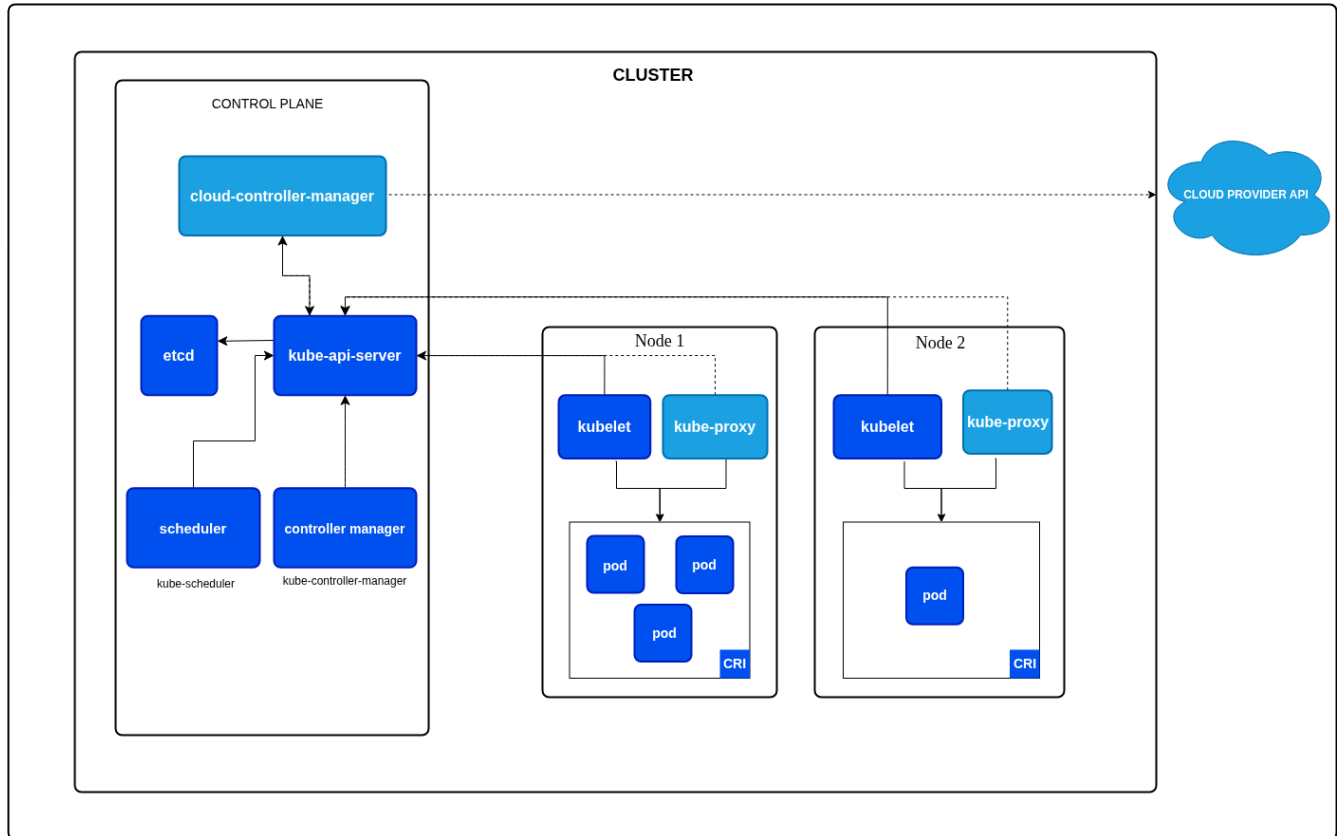
Screenshot of ConfigMap/Secret

```
horrondor@horrondor-Nitro-AN515-55:~$ kubectl get configmap
NAME                DATA  AGE
kube-root-ca.crt    1      45m
webapp-config        2      42m
horrondor@horrondor-Nitro-AN515-55:~$ kubectl get secrets
NAME                TYPE        DATA  AGE
webapp-secret        Opaque      2      35m
```

Screenshot of kubectl commands

```
297 kubectl cluster-info
298 kubectl get nodes
299 kubectl config get-contexts
301 kubectl config current-context
302 kubectl get nodes
304 kubectl get nodes
306 kubectl get nodes
307 kubectl get pods -o wide
308 kubectl get pods
309 kubectl get deployment
310 kubectl get services
311 kubectl get configmap
312 kubectl get secrets
314 kubectl get rs
315 kubectl scale deployment webapp --replicas=5
316 kubectl get pods -o wide
317 kubectl get deployment webapp -o wide
318 kubectl set image deployment/webapp nginx=nginx:1.28
319 kubectl rollout status deployment/webapp
320 kubectl get pods -w
321 kubectl rollout history deployment/webapp
322 kubectl rollout undo deployment/webapp
323 kubectl rollout status deployment/webapp
```

Kubernetes Architecture Diagram



Screenshot of 3 Running Replicas

```
horrondor@horrondor-Nitro-AN515-55:~$ kubectl get rs
NAME                                DESIRED    CURRENT    READY    AGE
webapp-5ffd64b6bb                   3          3          3        49m
horrondor@horrondor-Nitro-AN515-55:~$
```

Screenshot of Scaling

```
horrondor@horrondor-Nitro-AN515-55:~$ kubectl scale deployment webapp --replicas=5
kubectl get pods -o wide
deployment.apps/webapp scaled
```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
webapp-5ffd64b6bb-2n2l6	1/1	Running	0	43m	10.244.0.9	devops-control-plane	<none>	<none>
webapp-5ffd64b6bb-6rdf5	1/1	Running	0	52m	10.244.0.5	devops-control-plane	<none>	<none>
webapp-5ffd64b6bb-76c6x	0/1	ContainerCreating	0	0s	<none>	devops-control-plane	<none>	<none>
webapp-5ffd64b6bb-gw4qx	1/1	Running	0	52m	10.244.0.7	devops-control-plane	<none>	<none>
webapp-5ffd64b6bb-nx78r	0/1	ContainerCreating	0	0s	<none>	devops-control-plane	<none>	<none>

Screenshot of Rolling Update

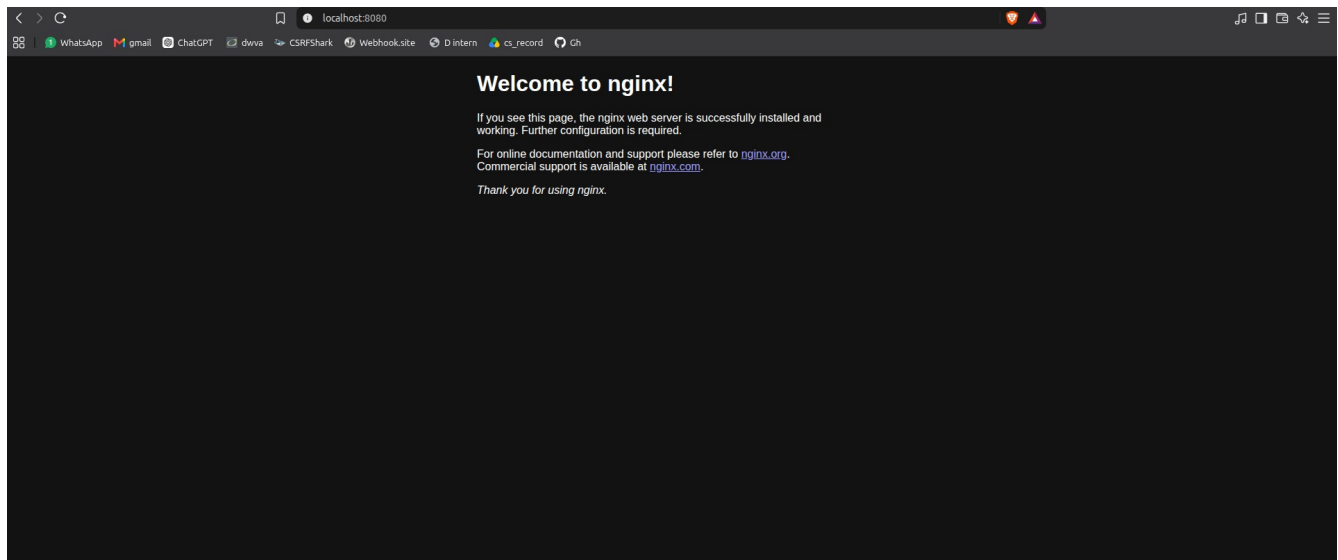
```
horrondor@horrondor-Nitro-AN515-55:~$ kubectl get deployment webapp -o wide
NAME      READY   UP-TO-DATE   AVAILABLE   AGE    CONTAINERS   IMAGES      SELECTOR
webapp    5/5     5            5           53m    nginx        nginx:1.27  app=webapp
horrondor@horrondor-Nitro-AN515-55:~$ kubectl set image deployment/webapp nginx=nginx:1.28
deployment.apps/webapp image updated
horrondor@horrondor-Nitro-AN515-55:~$ kubectl rollout status deployment/webapp
Waiting for deployment "webapp" rollout to finish: 3 out of 5 new replicas have been updated...
Waiting for deployment "webapp" rollout to finish: 3 out of 5 new replicas have been updated...
Waiting for deployment "webapp" rollout to finish: 3 out of 5 new replicas have been updated...
Waiting for deployment "webapp" rollout to finish: 3 out of 5 new replicas have been updated...
Waiting for deployment "webapp" rollout to finish: 4 out of 5 new replicas have been updated...
Waiting for deployment "webapp" rollout to finish: 4 out of 5 new replicas have been updated...
Waiting for deployment "webapp" rollout to finish: 4 out of 5 new replicas have been updated...
Waiting for deployment "webapp" rollout to finish: 4 out of 5 new replicas have been updated...
Waiting for deployment "webapp" rollout to finish: 2 old replicas are pending termination...
Waiting for deployment "webapp" rollout to finish: 2 old replicas are pending termination...
Waiting for deployment "webapp" rollout to finish: 2 old replicas are pending termination...
Waiting for deployment "webapp" rollout to finish: 2 old replicas are pending termination...
Waiting for deployment "webapp" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "webapp" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "webapp" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "webapp" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "webapp" rollout to finish: 4 of 5 updated replicas are available...
Waiting for deployment "webapp" rollout to finish: 4 of 5 updated replicas are available...
deployment "webapp" successfully rolled out
```

Screenshot of Rollback

```
horrondor@horrondor-Nitro-AN515-55:~$ kubectl rollout history deployment/webapp
deployment.apps/webapp
REVISION  CHANGE-CAUSE
1          <none>
2          <none>

horrondor@horrondor-Nitro-AN515-55:~$ kubectl rollout undo deployment/webapp
Warning: resource deployments/webapp was previously managed with 'kubectl apply'. Rolling back will not update the kubectl.kubernetes.io/last-applied-configuration annotation, which may cause unexpected behavior on future 'kubectl apply' operations. Consider using 'kubectl apply' with your previous configuration file instead.
deployment.apps/webapp rolled back
horrondor@horrondor-Nitro-AN515-55:~$ kubectl rollout status deployment/webapp
deployment "webapp" successfully rolled out
```

Screenshot of Application Access



Short explanation of what happened during scaling and rollback

Scaling

The Deployment was initially configured with 3 replicas. I scaled the Deployment up to 5 replicas, and Kubernetes automatically created additional Pods to match the desired replica count. I then scaled it back down to 3 replicas, and Kubernetes removed the extra Pods.

Rollback

I performed a rolling update by changing the application's container image. Kubernetes created a new Deployment revision and gradually replaced the old Pods. After verifying the rollout, I used `kubectl rollout undo` to roll back the Deployment to the previous revision. Kubernetes then restored the previous application version and successfully rolled out the Pods.